Homework-2026-09-29

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Screenshot:

```
➜  02 python3 validator.py data.xml schema.dtd
[VALID] data.xml conforms to schema.dtd.
➜  02 python3 validator.py invalid_data.xml schema.dtd
[INVALID] invalid_data.xml does not conform to schema.dtd.
  Line 4, column 0, Value "unknown" for attribute status of student is not among the enumerated set
```

```
1   <?xml version="1.0" encoding="UTF-8"?>
2   <!DOCTYPE students SYSTEM "schema.dtd">
3   <students>
4       <student id="S001" status="active">
```

```
1   <?xml version="1.0" encoding="UTF-8"?>
2   <!DOCTYPE students SYSTEM "schema.dtd">
3   <students>
4       <student id="S001" status="unknown">
```

Test results:

1. The valid XML document successfully conforms to the rules defined in schema.dtd.
2. The invalid test uses status="unknown", while the DTD only allows active or inactive. Therefore, the validator correctly detects the error and reports its location.

Conclusion:

The assignment successfully demonstrates XML and DTD validation using Python. The program can identify both valid and invalid XML documents and provides detailed validation errors, including line numbers and error descriptions.